

Nautilus Protocol

Whitepaper

Fibonacci controls supply. The market controls price. Nobody controls anything else.

Just math.

1. Introduction

Meme coins are fun.

The possibility of a small bet turning into a large return, the feeling of strangers gathering around the same coin — there is something genuinely exciting about it.

At the same time, a few structural challenges are well known. Sniper bots that buy in large quantities immediately after launch. Coordinated groups that accumulate early supply. Large sell orders that move the price significantly for everyone else.

These problems are difficult to solve completely. But it may be possible to improve things somewhat through design.

Nautilus is one such attempt.

By following the Fibonacci sequence for both price and supply, and holding sale proceeds in a program-derived address, the protocol aims to guarantee certain properties mathematically.

2. Design

Treasury

At the center of Nautilus is the treasury.

When tokens are purchased, SOL enters the treasury. When tokens are sold, SOL leaves the treasury.

The treasury is managed by a program-derived address (PDA). No private key exists for this address. SOL can only leave through the sell instruction, which requires burning tokens. Nobody — including the deployer — can withdraw treasury funds outside of the protocol.

The sell price is the weighted average of the treasury balance divided by tokens in circulation.

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sell price = treasury balance ÷ tokens in circulation
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As a result, market cap (MC) always equals the treasury balance.

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MC = treasury balance = SOL actually locked in the contract
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MC is not just a number. It has substance.

Fibonacci Price and Supply

Buy price and supply per stage both follow the Fibonacci sequence.

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sequence: 1, 1, 2, 3, 5, 8, 13, 21, 34, 55 ...
buy price = BASE_PRICE × FIB[stage]
supply    = FIB[stage] × 1,000,000 tokens
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BASE_PRICE is 0.001 SOL (1,000,000 lamports).

Stage 1 buy price is 0.001 SOL with a supply of 1,000,000 tokens. Stage 5 buy price is 0.005 SOL with a supply of 5,000,000 tokens. Price and supply increase together.

The stepped price structure raises the cost of large early purchases. Buying out the entire Stage 1 supply requires 1,000 SOL (approximately \$100,000). This makes early bulk accumulation economically costly.

A stage advances automatically when its supply is exhausted. Advancement is irreversible.

No On-chain Admin

After initialization, the protocol has no admin functions. There are no instructions to change prices, increase supply, or manipulate the treasury balance.

The protocol operates autonomously after deployment.

3. Two Core Properties

Property 1: The Treasury Cannot Be Drained

No matter how much selling occurs, the treasury balance cannot go negative.

Let T = treasury balance, N = tokens in circulation, k = tokens being sold.
After a sell:

$$\text{treasury_after} = T \times (1 - 0.995k/N)$$

As long as $k < N$ (at least one token remains), treasury_after is always positive.

This is a mathematical guarantee independent of market conditions. Regardless of SOL price movements or the volume of selling, the last token can always be sold.

Property 2: A Floor Price Exists

Nautilus has a floor price for your tokens.

The sell price is the weighted average of treasury balance divided by tokens in circulation. The treasury balance carries the memory of all past purchases. When someone sells a large amount, the treasury does not decrease proportionally. As tokens are removed from circulation, the treasury balance per remaining token stays the same or increases.

Even after a large sell, your token's floor price is maintained.

This is because of the treasury balance — a real asset that backs every token. In ordinary tokens, large sells push price down directly. In Nautilus, the sell price formula absorbs that impact.

Additionally, a 0.5% spread applies to all sells. The seller receives 99.5% of their proportional share; the remaining 0.5% stays in the treasury. As this spread accumulates, the floor price tends to rise slightly with each transaction.

4. The Spread Between Buy and Sell Price

In Nautilus, buy price and sell price diverge. Buy price follows the Fibonacci sequence and rises sharply. Sell price follows a weighted average and rises gradually.

When you calculate the ratio of buy to sell price across stages, it converges

toward the golden ratio ϕ (approximately 1.618).

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Stage 8:  buy/sell = 1.588
Stage 12: buy/sell = 1.614
Stage 20: buy/sell = 1.618  $\approx \phi$ 
```

This was not designed. It emerges naturally from two mathematical identities of the Fibonacci sequence.

$$\sum (FIB[i]^2) = FIB[n] \times FIB[n+1]$$
$$\sum (FIB[i]) = FIB[n+2] - 1$$

By choosing the Fibonacci sequence, mathematical properties known for over 2,000 years appeared in the protocol's price structure.

This spread is not merely a fee. The excess that later participants pay above the current weighted average accumulates in the treasury, supporting the floor price for all existing holders. The spread is what makes the floor possible.

5. Implementation

Nautilus is implemented as an Anchor program on Solana.

Instructions:

initialize — creates state account, SPL mint, and treasury PDA

buy(amount) — transfers SOL to treasury and mints tokens

sell(amount) — burns tokens and transfers SOL from treasury PDA

get_state — read-only query of current stage, prices, and treasury balance

A CLI tool is included for interacting with deployed instances via status, buy, sell, and balance commands.

In v0.4, upgrade authority is held by the deployer. In v0.5, upgrade authority will be revoked and the program will become immutable. The current authority status can be verified on-chain:

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solana program show <PROGRAM_ID>
```

6. Closing

Nautilus does not claim to solve all problems with meme coins.
It cannot eliminate snipers entirely, nor can it stop speculative behavior.
But a few things can be guaranteed mathematically.

No private key exists for the treasury.
No admin functions exist on-chain.
The treasury cannot be drained.
A floor price exists.
The buy/sell ratio converges to the golden ratio.

All of this is published as open-source code and can be verified by anyone.
The Fibonacci sequence was chosen. Mathematical properties known for over 2,000 years determined everything else.

Fibonacci controls supply. The market controls price. Nobody controls anything else. Just math.

Appendix: Price, MC, and Return by Stage

The table below shows reference values for each stage, assuming Stage 12 completion as the exit point.

- * Assumes SOL = \$100. Actual values vary with SOL price. No returns are guaranteed.
- * Multiples are calculated using the Stage 12 completion sell price (0.0888 SOL) as exit.
- * Stage 12 entry results in a loss at this exit assumption.

Stage	FIB	Supply	Buy (SOL)	Buy (USD)	MC at completion	Multiple	P&L
1	1	1,000,000	0.0010	\$0.10	\$100K	88.8x	+8,779%
2	1	1,000,000	0.0010	\$0.10	\$200K	88.8x	+8,779%
3	2	2,000,000	0.0020	\$0.20	\$600K	44.4x	+4,339%
4	3	3,000,000	0.0030	\$0.30	\$2M	29.6x	+2,860%
5	5	5,000,000	0.0050	\$0.50	\$4M	17.8x	+1,676%
6	8	8,000,000	0.0080	\$0.80	\$10M	11.1x	+1,010%

7	13	13,000,000	0.0130	\$1.30	\$27M	6.8x	+583%
8	21	21,000,000	0.0210	\$2.10	\$71M	4.2x	+323%
9	34	34,000,000	0.0340	\$3.40	\$187M	2.6x	+161%
10	55	55,000,000	0.0550	\$5.50	\$490M	1.6x	+61%
11	89	89,000,000	0.0890	\$8.90	\$1.28B	1.0x	±0%
12	144	144,000,000	0.1440	\$14.40	\$3.36B	0.6x	-38%

MC at completion = treasury balance at the point when all stages up to that stage are sold out.

Stage 11 is approximately break-even. Stage 12 entry results in a loss.

This document is for informational purposes only. It does not constitute financial advice or an offer to purchase any token.

Nautilus is open-source software. The code is publicly available.